



# Madhan Rasipuram Ravichandran

MODEL-BASED DEVELOPMENT ENGINEER

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## Professional Summary

Model-Based Development Engineer with over 3 years of experience in automotive software development, specializing in Battery Management Systems and Fuel Cell technologies. Strong expertise in MATLAB/Simulink, Stateflow, Embedded C, and Model-Based Design workflows. Proven experience working in OEM and Tier-1 environments with strict adherence to MISRA and AUTOSAR standards. Effective in requirements implementation, model optimization, testing, and production-quality code generation.

Highly skilled in MATLAB/Simulink to implement complex logics. Familiar with Bazel framework for testing C codes and YAML configuration for GitHub Action workflows. Hands-on experience with PC based oscilloscope.

A frequent contributor in MATLAB Answers and a **MVP** and **Masters badge** holder. Absolute fan of MathWorks. A curious individual up for challenges and to learn new concepts. Strongly believes Model

Based Design (MBD) and Model Based Systems Engineering (MBSE) makes everything much easier and simpler.

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## Skills & Expertise

### Automotive & Embedded Systems

Bare metal (direct register level programming) · Infineon AURIX TC375 Tricore · Aurix TC375 Tricore Assembly · miniMCDS On-Chip Memory Tracing · STM32 · STM32G431(R/K)B · STM32H7(45/55)ZI-Q · STM32H735G-DK · Embedded C · C · MISRA C · Automotive SPICE (ASPICE) · EOL Testing

### Communication Protocols

Controller Area Network (CAN) · Controller Area Network Flexible Data-Rate (FDCAN) · Unified Diagnostic Services over Controller Area Network (UDSonCAN) · On-Board Diagnostic2 over Controller Area Network (OBD2onCAN) · Unified Diagnostic Services (UDS) · On-board diagnostics2 (OBD2) · Serial Peripheral Interface (SPI) · Inter-Integrated Circuit (I2C) · Universal Asynchronous Receiver/Transmitter (UART/USART) · USB CDC

### MBD/MBSE & Simulation

MATLAB · Simulink · Stateflow · Simulink Coverage · MATLAB App Designer · MATLAB Embedded Coder · Model-Based Design (MBD) · Model-Based Systems Engineering (MBSE) · MathWorks Profiler · PSIM

### Testing & Measurement

MIL / SIL / HIL Testing · Picoscope 2208B MSO

### CI/CD & Development Tools

Git · GitHub Actions · Bazel · Jira · SonarQube · YAML · JSON · Microsoft Visual Studio Code · Eclipse

### Systems Engineering & Design Tools

Sparx Enterprise Architect · Rational DOORS · PTC Integrity · Printed Circuit Board (PCB) Design · KiCad · SolidWorks

### Engineering Software & Analysis

ANSYS · LaTeX · Python · PowerShell · JavaScript · HTML · Apple Shortcuts · Digitale Signalprozessoren · Adaptive Signal Processing · Image Processing · Microsoft Excel · Microsoft Word

### Domain Knowledge

Battery Management Systems (BMS) · Batteriemanagementsysteme · Fuel Cell Systems · Power Electronics · Electric Motor Controls · ISO 26262 · PDM

## Languages

English C1 – Professional · German B2 – Professional

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## Professional Experience

### Model Development Engineer - BMS

**Capgemini Engineering (for BMW Group AG)** – Munich, Germany *July 2023 – October 2025*

- Developed charging management algorithms for Battery Management Systems using MATLAB/Simulink
  - Implemented functional requirements in collaboration with functional and system developers
  - Executed Model-in-the-Loop and Software-in-the-Loop test runs and generated automated test reports using GitHub Actions
  - Generated Embedded C code using MATLAB Embedded Coder with Bazel framework integration
  - Optimized model logic and resolved defects identified during testing phases
  - Performed Model-in-the-Loop, Software-in-the-Loop, and Hardware-in-the-Loop validation
  - Ensured compliance with MISRA and AUTOSAR modeling and coding standards
  - Conducted static code analysis using SonarQube
  - Analyzed HIL-level defects down to low-level artifacts (.c, .a2l, .slx)
  - Reviewed peer pull requests and supported continuous integration workflows
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### Fuel Cell Development Engineer

**AVL** – Warsaw, Poland *December 2022 – May 2023*

- Developed and validated fuel cell system models
  - Performed system-level simulations using MATLAB/Simulink
  - Supported testing, analysis, and validation activities for fuel cell powertrain components
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### Fuel Cell Intern

**AVL** – Warsaw, Poland *August 2022 – November 2022*

- Assisted in fuel cell system modeling and simulation tasks
  - Supported senior engineers with data analysis and validation activities
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### Virtual C-Ops Associate

**Amazon** – Remote *July 2021 – July 2022*

- Executed operational support tasks in a virtual environment
- Ensured process compliance, data accuracy, and service quality

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## Education

### Bachelor of Engineering - Electric and Hybrid Vehicle Engineering

Warsaw University of Technology 2016 - 2022 | Grade: **4.42**

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## Licenses & Certifications

### Certificate WBT SEPA M1 - System Engineering Fundamentals

AVL | Issued December 2022

### Stateflow Onramp

MathWorks | Issued November 2022

### GOETHE-ZERTIFIKAT B2

Goethe-Institut e.V. | Issued February 2022 | Skills: Deutsch

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## Personal Projects & Adventures

### GitHub Actions CI Pipeline for Infineon AURIX TC375

- Built a GitHub Actions CI pipeline for Infineon AURIX TC375 firmware development using GCC and Memory Utilization Dashboard **from scratch**
- Validates builds automatically on each commit and generates firmware artifacts including **ELF, HEX, MAP, LST**, and size reports
- Achieved reproducible builds across local and cloud environments, handling artifact retention and daily Markdown report generation
- Powershell script for fetching latest builds and flashing it in Infineon AURIX TC375 using **Aurix Flasher**
- Automated HTML report generation through MATLAB using GitHub API methods and PowerShell scripts in CI
- Created a `MATLABDiffGenerator` to produce extended inline diffs that go beyond the default GitHub diff view
- Added a live view grapher custom JavaScript function in the existing Aurix TCF VSCode Debugger for better understanding of system behavior
- Utilized **Apple Shortcuts** to trigger workflow dispatch for higher precision than cron schedulers, and to convert Markdown files to CSS-styled HTML for mobile readability

- *Only for Personal Use.*

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## Variable PWM using GTM Module TOM - Infineon AURIX TC375

Implemented **bare metal register level programming** to perform Variable PWM using GTM Module TOM in Infineon AURIX Tricore 375 and validated behaviour using **Picoscope 2208B MSO**.

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## ISR Execution & Latency Measurement - AURIX TC375

Measured AURIX Tricore TC375 ISR execution and ISR Latency using **CPU Performance Counter** and validated it through hardware GPIO measurement using **Picoscope 2208B MSO**.

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## GCC vs Tasking Compiler Assembly Analysis - AURIX TC3xx

Analyzed AURIX Tricore TC3xx **GCC Compiler vs Tasking Assembly** with same and different optimization flags, thereby understanding the AURIX Tricore TC375's architecture very well.

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## STM32 Embedded Systems & Communication Protocols

- Hands-on with **Picoscope 2208B MSO** and **STM32G431(R/K)B / STM32H7(45/55)ZI-Q / STM32H735G-DK** using HAL Libraries and **pure direct register level programming**
  - Implemented **FD/CAN, SPI, I2C, USB CDC, and USART** communication protocols successfully and validated using Picoscope
  - Implemented indicators on **LCD TFT of STM32H735G-DK** for crucial data display
  - Performed **UDS on CAN** on real vehicle using STM32 (*OBD2 on CAN - yet to be done*)
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## Vehicle CAN Reverse Engineering & Logging

- Developed **MATLAB application** which logs vehicle's CAN Frames, plots bytes using CAN ID in real time, visualizes all CAN IDs bytes using STM32H735G-DK
- Successfully **reverse engineered CAR CAN IDs** - identifying what each CAN ID and its bytes represent
- Used **C for logging** to reduce overhead and capture all CAN Frames at ~150 microseconds intervals
- Designed a **PCB** to interface STM32G431KB with OBD2 port

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## Community Contributions

- [MATLAB Answers - Profile](#) - Frequent contributor, **MVP** and **Masters badge** holder
- A blog about me due to highest contributions in MATLAB Answers: [Community Q&A - Madhan Ravi](#)
- Passionate advocate of MathWorks ecosystem and Model-Based Design methodology

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